SDM COLLEGE OF ENGINEERING AND TECHNOLOGY,

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(An autonomous Institution affiliated to
Visvesvaraya Technological University, Belgaum – 590018)

Department of Electronics and Communication Engineering

Minor project-2 [18UECL604] Report entitled

"BRAIN TUMOR SEGMENTATION USING IMAGE PROCESSING"

Submitted by

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Students of 6^{th} Semester
Under the guidance of
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Department of Electronics and Communication Engineering
Academic Year 2022-23

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CERTIFICATE

This is to certify that, **Mr. Ajay M.Rathod(2SD20EC005),Ms. Akshata S.Katrale(2SD20EC010),Ms. Inchara Nayaka(2SD20EC043),Mr. Mohammed Yaseen(2SD20EC057)** students of 6th Semester have satisfactorily completed the **Minor Project-2** entitled **"Brain tumor segmentation using image processing".** The report has been examined and is approved that it satisfies the necessary academic requirements. The report is submitted to the **Department of Electronics and Communication Engineering, SDM College of Engineering and Technology, Dharwad – 580 002** during the academic year 2022-23.

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Examiner – 1	**Examiner – 2**
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ACKNOWLEDGEMENT

We consider it a privilege to express our sincere gratitude and respect to all those who have guided and inspired us for this project.

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ABSTRACT

Now a day's tumor is second leading cause of cancer. Due to cancer large no of patients are in danger. The medical field needs fast, automated, efficient and reliable technique to detect tumor like brain tumor. Detection plays very important role in treatment. If proper detection of tumor is possible then doctors keep a patient out of danger. Various image processing techniques are used in this application. Using this application doctors provide proper treatment and save a number of tumor patients. A tumor is nothing but excess cells growing in an uncontrolled manner. Brain tumor cells grow in a way that they eventually take up all the nutrients meant for the healthy cells and tissues, which results in brain failure. Currently, doctors locate the position and the area of brain tumor by looking at the MR Images of the brain of the patient manually. This results in inaccurate detection of the tumor and is considered very time consuming. A tumor is a mass of tissue it grows out of control. We can use a Segmentation Method by Image Processing.

LIST OF FIGURES

LIST OF ABBREVATIONS

Acronym	Explication
MRI	Magnetic resonance imaging
CNN	Convolutional neural network

PROJECT REPORT ORGANIZATION

Chapter 1 Presents introduction to the overall thesis and the overview of the project. In the project overview a brief introduction of Brain Tumor detection using image processing and its applications and advantages are discussed.

Chapter 2 Presents the literature survey which has research papers of the projects. Related background and concept , problem statement and block diagram which explains the essence of two technologies that we use in this project.

Chapter 3 Stages of development, Flowcharts, Methodologies and Techniques to be used, Software tools

Chapter 4 This chapter shows the outcomes of the project.

Chapter 5 Explains the , future scope, conclusion.

Chapter 6 Explain the references

CHAPTER -1

1.1 INTRODUCTION

The human body is made up of many organs and brain is the most critical and vital organ of them all. One of the common reasons for dysfunction of brain is brain tumor. A tumor is nothing but excess cells growing in an uncontrolled manner. Brain tumor cells grow in a way that they eventually take up all the nutrients meant for the healthy cells and tissues, which results in brain failure. Currently, doctors locate the position and the area of brain tumor by looking at the MRI Images of the brain of the patient manually. This results in inaccurate detection of the tumor and is considered very time consuming.

A Brain Cancer is very critical disease which causes deaths of many individuals. The brain tumor detection and classification system is available so that it can be diagnosed at early stages. Cancer classification is the most challenging tasks in clinical diagnosis.

This project deals with such a system, which uses computer, based procedures to detect tumor blocks and classify the type of tumor using Convolution Neural Network Algorithm for MRI images of different patients. Different types of image processing techniques like image segmentation, image enhancement and feature extraction are used for the brain tumor detection in the MRI images of the cancer-affected patients. Detecting Brain tumor using Image Processing techniques its involves the four stages is Image Pre-Processing, Image segmentation, Feature Extraction, and Classification. Image processing and neural network techniques are used for improve the performance of detecting and classifying brain tumor in MRI images.

1.2 MOTIVATION

The main motivation behind Brain tumor detection is to not only detect tumor but it can also classify types of tumor. So it can be useful in cases such as we have to sure the tumor is positive or negative, it can detect tumor from image and return the result tumor is positive or not. This project deals with such a system, which uses computer, based procedures to detect tumor blocks and classify the type of tumor using Convolution Neural Network Algorithm for MRI images of different patients.

1.3 APPLICATIONS

- The main aim of the applications is tumor identification.
- The main reason behind the development of this application is to provide proper treatment as soon as possible and protect the human life which is in danger.
- This application is helpful to doctors as well as patient.
- The manual identification is not so fast, more accurate and efficient for user. To overcome those problem this application is design.
- It is user friendly application.

CHAPTER-2

PROJECT MODEL

2.1 Literature Survey

Automated Brain Tumor Segmentation Using Hybrid Image Processing Techniques" by B.S. Swathi and S.K. Somasundaram:

This paper proposes a hybrid approach that combines the K-means clustering algorithm and the watershed transform for brain tumor segmentation. The proposed method achieves high accuracy in segmenting both low-grade and high-grade brain tumors.

"Automated Brain Tumor Segmentation Using Convolutional Neural Networks" by H. Chen, Q. Dou, L. Yu:

This paper proposes a deep learning-based method for brain tumor segmentation that uses convolutional neural networks (CNNs). The proposed method achieved high accuracy in segmenting gliomas and meningiomas, and outperformed several other state-of-the-art methods.

"Segmentation of Brain Tumors in MRI Images Using Differential Evolution" by M.A. Siddiqui and M.A. Imran:

This paper proposes a hybrid method that combines fuzzy c-means clustering and differential evolution for brain tumor segmentation. The proposed method achieved high accuracy in segmenting gliomas and meningiomas.

"A Comparative Study of Image Segmentation Algorithms for Brain Tumor Detection" by H.A. Jalab, A.H. Hassan, and K.S. Kareem:

This paper presents a comparative study of various image segmentation algorithms for brain tumor detection, including region growing, thresholding, watershed transform, and active contour models. The study found that the active contour model achieved the highest accuracy in segmenting brain tumors.

"Brain Tumor Segmentation Using Adaptive Thresholding and Morphological Operations" by S. Sujatha and S. Karpagavalli:

This paper proposes a method for brain tumor segmentation that uses adaptive thresholding and morphological operations. The proposed method achieved high accuracy in segmenting both low-grade and high-grade brain tumors.

2.2 PROBLEM STATEMENT

To detect the Brain Tumor and Segment the image using MATLAB Code.

CHAPTER-3

3.1 DATA FLOW DIAGRAM

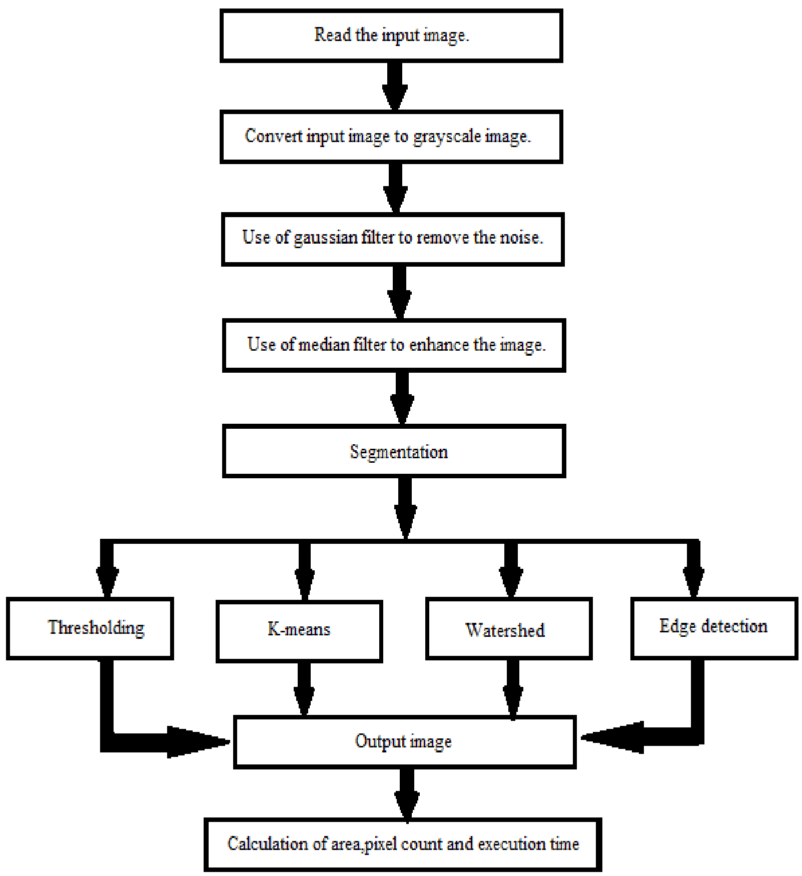

FIGURE 1: Work flow graph of brain tumor detection

3.2 METHODOLOGY

3.2.1 MRI images of the brain

This is the first step of the proposed system. The resulting MRI images may not be of very good quality for analysis. Images can be noisy, blurry, low-contrast. The area of interest can be difficult to extract [14]. Here, grayscale MRI images are provided as input to the system.

3.2.2 Convert it to gray scale image

This step helps in improving the Standard of the image, making it more suitable for further processing. Here, we convert the RGB input image to gray-scale. It also includes steps like Image enhancement, contrast improvement and image sharpening.

3.2.3 Apply high pass filter for noise removal

Noise removal is the important process of image processing. In this processing improving the image quality. So, the first step is to remove the noise present in the MRI image. The methods used for noise reduction can be linear or non-linear. In Linear filters, for noise reduction, the pixel value is updated by weighted average of neighbourhoods. This method reduces the image quality. On the other hand in non-linear method, the edges are preserved but the fine structures are degraded. Here, we are using a Gaussian blur filter and median filter. Gaussian blur filter is non linear filter, it blur the image and reduced the noise. The median filter is a nonlinear filtering, often used to remove noise from an image or signal.

3.2.4 Apply anisotropic filter to enhance the quality of image

To enhance the quality of an image in brain tumor segmentation using image processing, you can apply an anisotropic filter. Anisotropic filtering is a technique that selectively applies smoothing to an image, preserving edges and important features while reducing noise.

3.2.5 Compute threshold segmentation

Segmentation require separating an image into regions corresponding to objects. Threshold is the simplest method of segmenting image. Threshold can be used to convert Gray-Scale image to binary images. The simplest property that pixels in a pixels can share is intensity. So, a natural way to segment such pixels is through threshold, the separation of light and dark pixels. Threshold convert the gray images to binary image, its value to convert the zero's and one's. Threshold have a many types. One of the approach is OTSU threshold. Otsu's threshold technique includes iterating via all the feasible threshold values and calculating a measure of unfold for the pixel ranges every aspect of the threshold, i.e. the pixels that both fall in foreground or background. The intention is to discover the threshold value of foreground and background of the binary images.

3.2.6 Compute morphological operation

Morphological operations play an essential role in brain tumor segmentation using image processing. They are mathematical operations that manipulate the shape, size, and connectivity of objects in an image. In the context of brain tumor segmentation, morphological operations can be used to enhance or extract specific features related to tumor regions.

The two fundamental morphological operations used in tumor segmentation are erosion and dilation. These operations are typically applied in combination or sequentially to achieve the desired results.

1. Erosion:

Erosion is a morphological operation that shrinks or erodes the boundaries of objects in an image. It achieves this by scanning the image with a structuring element (a small matrix or kernel) and replacing each pixel with the minimum value within the neighborhood defined by the structuring element. Erosion helps to remove small details and noise while preserving the overall shape and connectivity of larger structures.

In brain tumor segmentation, erosion can be used to remove noise and small artifacts surrounding the tumor region, resulting in a smoother and more defined tumor boundary.

2. Dilation:

Dilation is the inverse of erosion and expands or dilates the boundaries of objects in an image. It replaces each pixel with the maximum value within the neighborhood defined by the structuring element. Dilation helps to fill in gaps, connect fragmented regions, and enlarge the objects.

In brain tumor segmentation, dilation can be used to bridge gaps within the tumor region and ensure a more complete and connected segmentation result.

By combining erosion and dilation, additional morphological operations can be performed:

3. Opening:

Opening is an operation that applies erosion followed by dilation. It is useful for removing small bright regions (e.g., noise) and filling in small dark holes within objects. Opening can help to refine the tumor segmentation by removing small artifacts and enhancing the tumor region's shape.

4. Closing:

Closing is an operation that applies dilation followed by erosion. It is effective in filling small dark regions (e.g., holes) and bridging small bright gaps between objects. Closing can help to close gaps within the tumor region and improve the overall segmentation accuracy.

These morphological operations are often combined with other image processing techniques, such as thresholding, region-growing, or connected component analysis, to achieve accurate and robust brain tumor segmentation. The choice of structuring element size and shape, as well as the sequence of operations, may vary depending on the specific characteristics of the tumor and the desired segmentation outcome.

It's important to note that morphological operations alone may not be sufficient for complete tumor segmentation. They are typically part of a larger pipeline that involves various preprocessing steps, feature extraction, and advanced algorithms to achieve accurate and reliable tumor segmentation results.

3.2.7 Finally output will be a tumor region

3.3 SOFTWARE:
MATLAB

CHAPTER 4

OUTCOMES

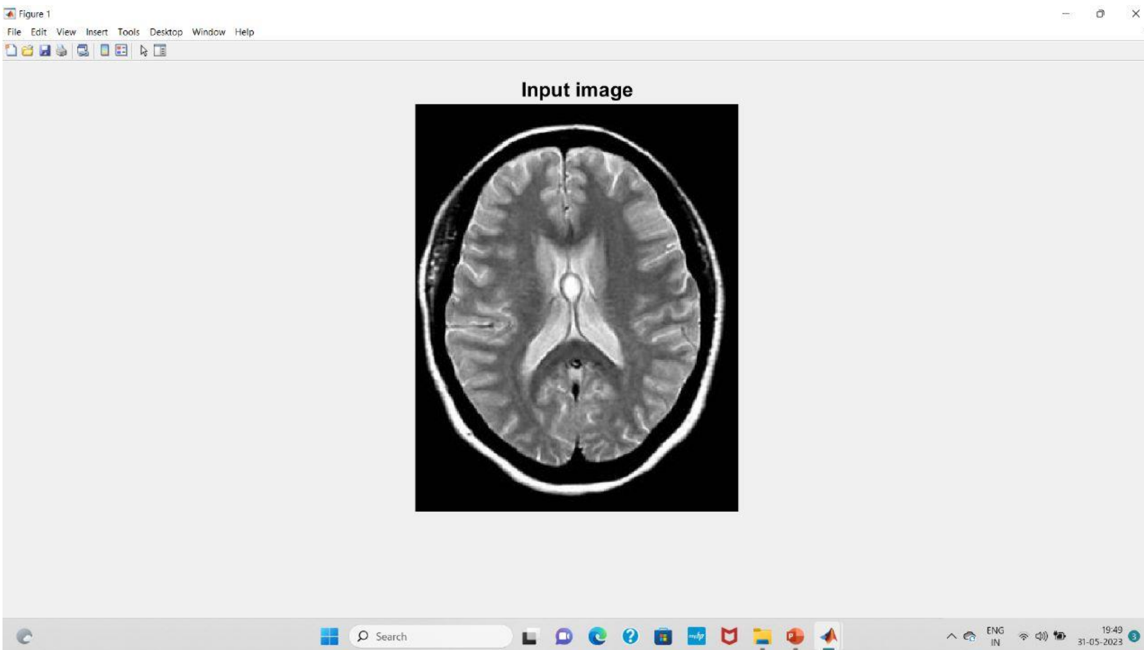

FIGURE 2: Input image

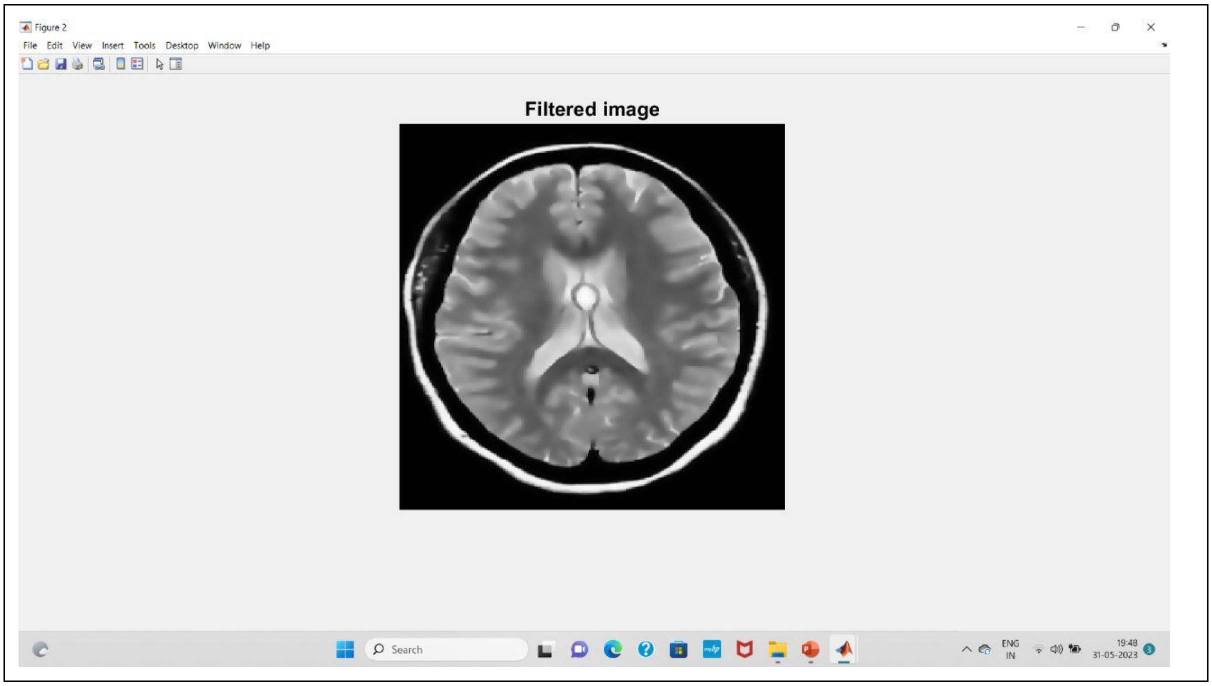

FIGURE 3: Filtered image

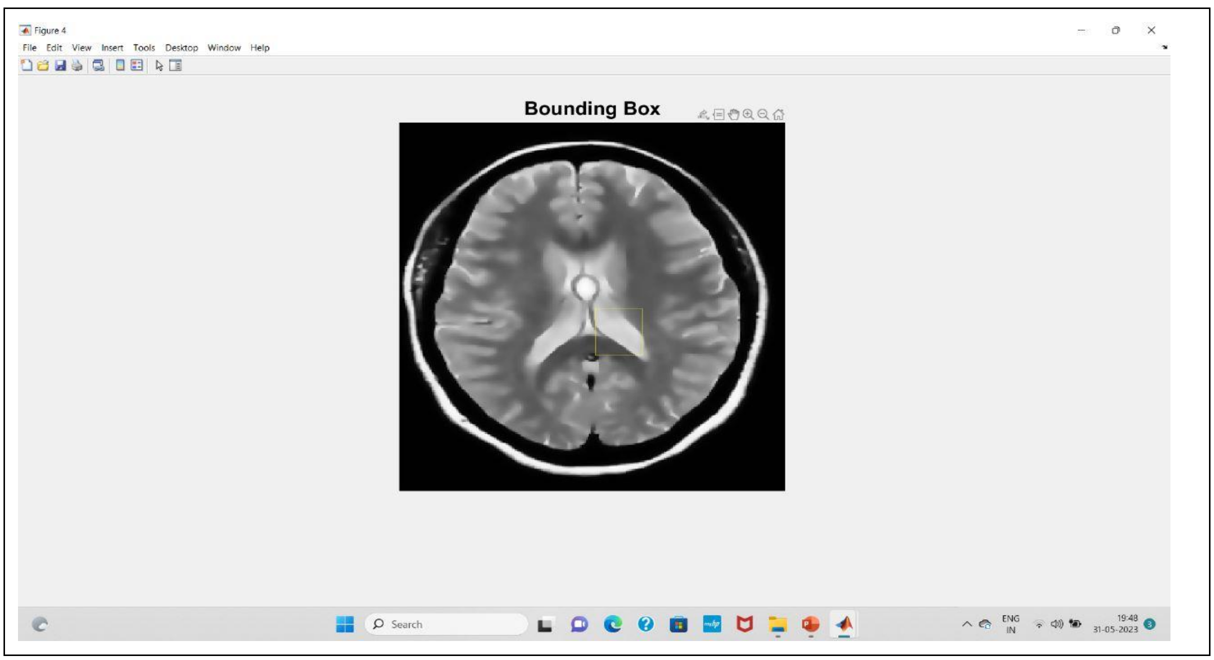

FIGURE 4: Bounding box

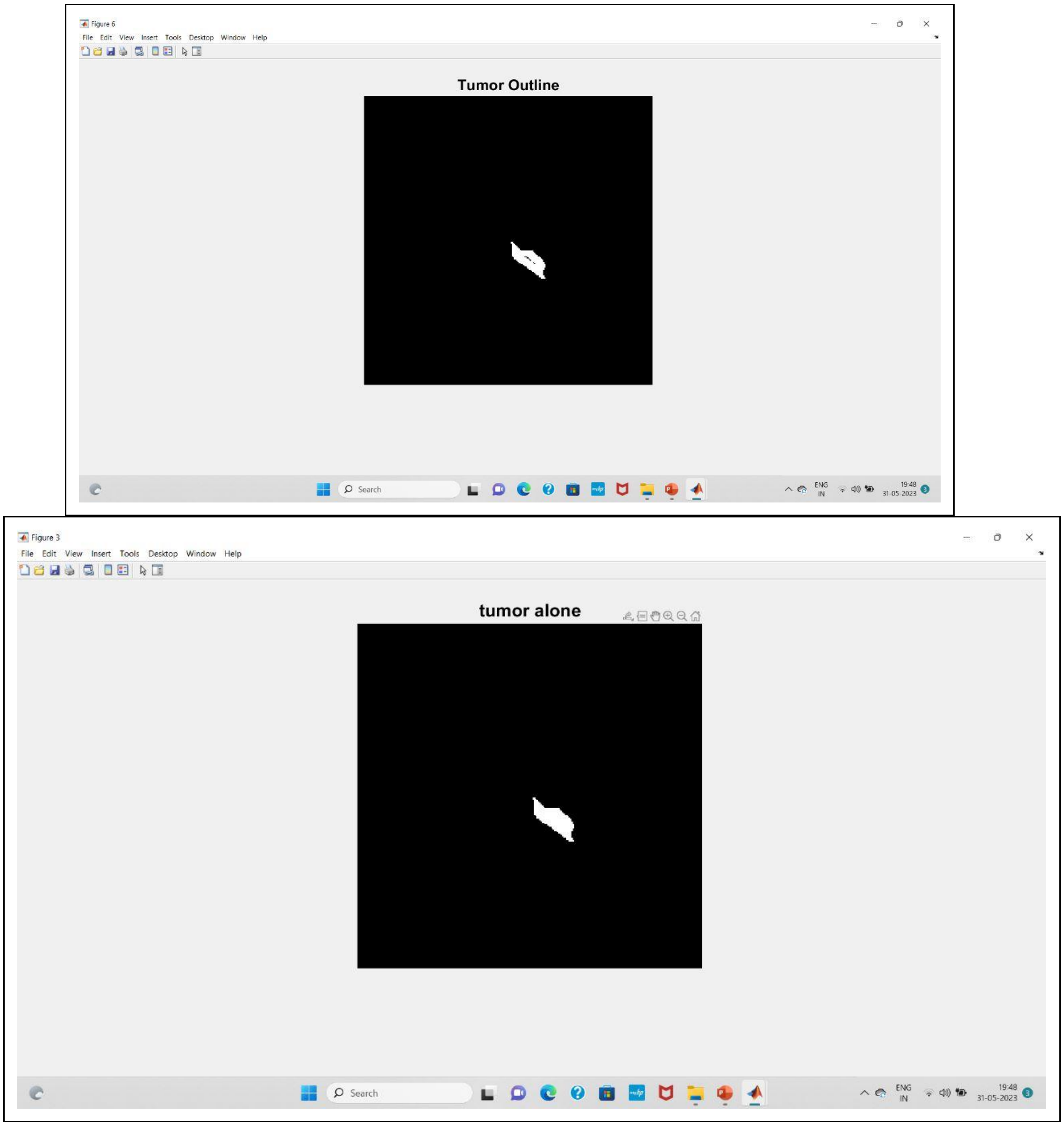

FIGURE 5: Tumor alone and outline

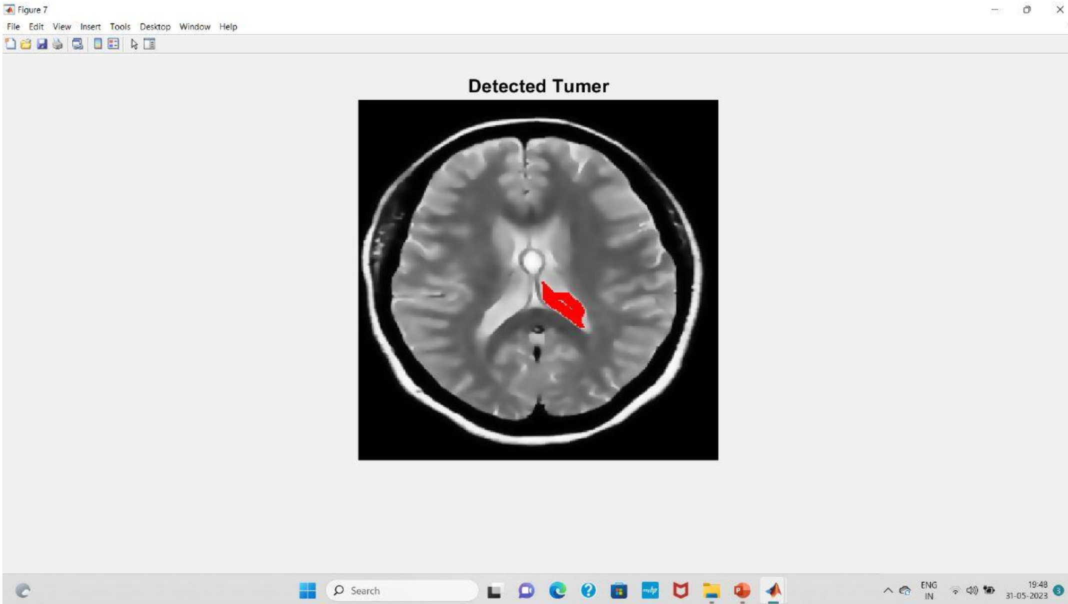

FIGURE 6 : Detected tumor

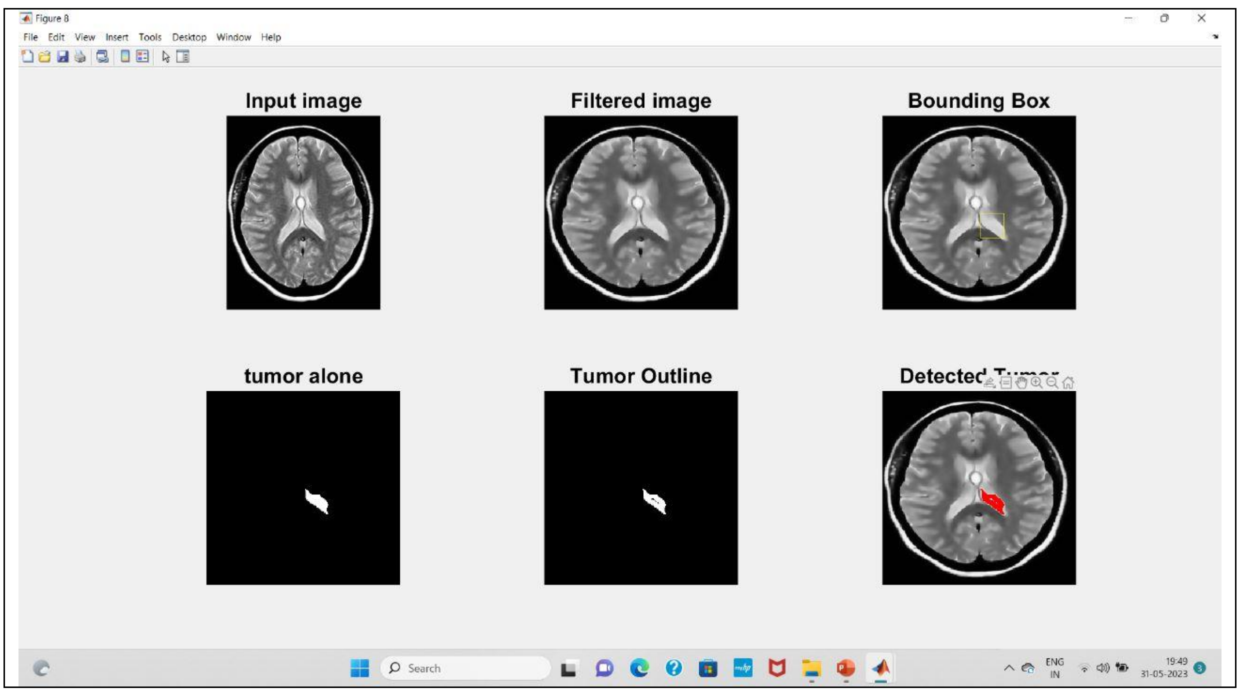

FIGURE 7: All images

CHAPTER 5

5.1 CONCLUSION

Brain tumor segmentation is a crucial task in medical imaging analysis, as it aids in accurate diagnosis, treatment planning, and monitoring of tumor progression. MATLAB, with its extensive image processing capabilities and toolboxes, provides a powerful platform for performing brain tumor segmentation. Through the use of MATLAB's image processing functions, researchers and clinicians can implement various segmentation algorithms to extract tumor regions from brain images. These algorithms can be broadly classified into two categories: supervised and unsupervised methods. Supervised methods require manual annotation of the tumor regions in a training dataset to train a segmentation model. Additionally, MATLAB offers various preprocessing techniques to enhance the quality of brain images before segmentation. These techniques include denoising, intensity normalization, bias correction, and image registration, which help improve the accuracy and robustness of tumor segmentation algorithms. By leveraging the capabilities of MATLAB, researchers and clinicians can evaluate and compare the performance of different segmentation methods using quantitative metrics such as sensitivity, specificity, dice coefficient, and Hausdorff distance. These evaluations help in selecting the most appropriate algorithm for a given dataset or clinical scenario. In conclusion, MATLAB provides a comprehensive platform for brain tumor segmentation. Its image processing functions, machine learning capabilities, and extensive toolboxes enable the development and implementation of both supervised and unsupervised segmentation methods. By utilizing MATLAB's capabilities, researchers and clinicians can enhance the accuracy of brain tumor segmentation, facilitating better diagnosis and treatment planning for patients with brain tumors.

5.2 FUTURE WORK

The possibilities for detecting a brain tumor in the future are that if we get a three-dimensional image of the brain with the tumor, then we can also estimate the type of tumor as well as the stage of the tumor.

CHAPTER 6

REFERENCES

1. **“Automated Brain Tumor Segmentation Using Hybrid Image Processing Techniques”** by B.S. Swathi and S.K. Somasundaram.

2. **"Automated Brain Tumor Segmentation Using Convolutional Neural Networks"** by H. Chen, Q. Dou, L. Yu.

3. **"Segmentation of Brain Tumors in MRI Images Using Differential Evolution"** by M.A. Siddiqui and M.A. Imran.

4. **"A Comparative Study of Image Segmentation Algorithms for Brain Tumor Detection"** by H.A. Jalab, A.H. Hassan, and K.S. Kareem.

5. **"Brain Tumor Segmentation Using Adaptive Thresholding and Morphological Operations**" by S. Sujatha and S. Karpagavalli.